C

IB MATH: APPLICATIONS & INTERPRETATION • UNIT 5.01

Distributions & Maximum Likelihood

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 A
Q6 A	Q7 A	Q8 A	Q9 B	Q10 C
Q11 D	Q12 C	Q13 A	Q14 A	Q15 D
Q16 B	Q17 B	Q18 C	Q19 A	Q20 C
Q21 A	Q22 B	Q23 B	Q24 C	Q25 A

Distributions & Maximum Likelihood Practice Exercises

1 MCQ

A feature in an AI vision model follows a normal distribution $X \sim \mathrm{N}(25, 36)$. What is the mean μ and variance σ^2 of this feature?

- **A** $\mu = 25,\ \sigma^2 = 36$ ✓
- **B** $\mu = 36,\ \sigma^2 = 25$
- **C** $\mu = 25,\ \sigma^2 = 6$
- **D** $\mu = 6,\ \sigma^2 = 36$

WORKED SOLUTION & VERIFICATION

The notation $\mathrm{N}(\mu, \sigma^2)$ directly specifies the mean $\mu = 25$ and variance $\sigma^2 = 36$ (with standard deviation $\sigma = 6$).

2 MCQ

Given $n = 4$ independent observations from a normal distribution with sum $\sum_{i=1}^{4} x_i = 12$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\mathrm{MLE}}$?

- **A** $\hat{\mu}_{\mathrm{MLE}} = 3$ ✓
- **B** $\hat{\mu}_{\mathrm{MLE}} = 6$
- **C** $\hat{\mu}_{\mathrm{MLE}} = 12$
- **D** $\hat{\mu}_{\mathrm{MLE}} = 1.5$

WORKED SOLUTION & VERIFICATION

The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{12}{4} = 3$.

3 MCQ

Given $n = 8$ independent observations from a normal distribution with sum $\sum_{i=1}^{8} x_i = 24$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

- **A** $\hat{\mu}_{\text{MLE}} = 3$ ✓
- **B** $\hat{\mu}_{\text{MLE}} = 6$
- **C** $\hat{\mu}_{\text{MLE}} = 24$
- **D** $\hat{\mu}_{\text{MLE}} = 1.5$

WORKED SOLUTION & VERIFICATION

The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{24}{8} = 3$.

4 MCQ

Given $n = 12$ independent observations from a normal distribution with sum $\sum_{i=1}^{12} x_i = 36$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

- **A** $\hat{\mu}_{\text{MLE}} = 3$ ✓
- **B** $\hat{\mu}_{\text{MLE}} = 6$
- **C** $\hat{\mu}_{\text{MLE}} = 36$
- **D** $\hat{\mu}_{\text{MLE}} = 1.5$

WORKED SOLUTION & VERIFICATION

The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{36}{12} = 3$.

5 MCQ

Given $n = 16$ independent observations from a normal distribution with sum $\sum_{i=1}^{16} x_i = 48$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

- **A** $\hat{\mu}_{\text{MLE}} = 3$ ✓
- **B** $\hat{\mu}_{\text{MLE}} = 6$
- **C** $\hat{\mu}_{\text{MLE}} = 48$
- **D** $\hat{\mu}_{\text{MLE}} = 1.5$

WORKED SOLUTION & VERIFICATION

The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{48}{16} = 3$.

6 MCQ

Given $n = 20$ independent observations from a normal distribution with sum $\sum_{i=1}^{20} x_i = 60$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

- **A** $\hat{\mu}_{\text{MLE}} = 3$ ✓
- **B** $\hat{\mu}_{\text{MLE}} = 6$
- **C** $\hat{\mu}_{\text{MLE}} = 60$
- **D** $\hat{\mu}_{\text{MLE}} = 1.5$

WORKED SOLUTION & VERIFICATION

The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{60}{20} = 3$.

7 MCQ

Given $n = 24$ independent observations from a normal distribution with sum $\sum_{i=1}^{24} x_i = 72$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

- **A** $\hat{\mu}_{\text{MLE}} = 3$ ✓
- **B** $\hat{\mu}_{\text{MLE}} = 6$
- **C** $\hat{\mu}_{\text{MLE}} = 72$
- **D** $\hat{\mu}_{\text{MLE}} = 1.5$

WORKED SOLUTION & VERIFICATION

The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{72}{24} = 3$.

8 MCQ

In training a language model token predictor, a binary token occurs $k = 3$ times across $n = 10$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

- **A** $\hat{p}_{\text{MLE}} = 0.300$ ✓
- **B** $\hat{p}_{\text{MLE}} = 0.450$
- **C** $\hat{p}_{\text{MLE}} = 0.150$
- **D** $\hat{p}_{\text{MLE}} = 0.700$

WORKED SOLUTION & VERIFICATION

For a Bernoulli likelihood $L(p) = p^k(1-p)^{n-k}$, the log-likelihood is $\ln L = k\ln p + (n-k)\ln(1-p)$. Setting derivative to zero: $\frac{k}{p} - \frac{n-k}{1-p} = 0 \implies \hat{p} = \frac{k}{n} = \frac{3}{10} = 0.300$.

9 MCQ

Consider a discrete random variable X with the following probability mass function (PMF): $P(X = x) = \frac{x}{10}$ for $x \in \{1, 2, 3, 4\}$. What is the value of $P(X \leq 2)$?

- **A** $\frac{1}{10}$
- **B** $\frac{3}{10}$ ✓
- **C** $\frac{6}{10}$
- **D** $\frac{4}{10}$

WORKED SOLUTION & VERIFICATION

The probability mass function is given by $P(X = x) = \frac{x}{10}$ for $x \in \{1, 2, 3, 4\}$. We need to find $P(X \leq 2)$. This is the sum of probabilities for $X = 1$ and $X = 2$. $P(X \leq 2) = P(X = 1) + P(X = 2)$. $P(X = 1) = \frac{1}{10}$ and $P(X = 2) = \frac{2}{10}$. Therefore, $P(X \leq 2) = \frac{1}{10} + \frac{2}{10} = \frac{3}{10}$.

10 MCQ

Principal Component Analysis on covariance matrix Σ_A yields eigenvalues $\lambda_1 = 51$ and $\lambda_2 = 49$. Determine the percentage of variance f_x explained by the first principal component.

- **A** 33
- **B** 55
- **C** 51 ✓
- **D** 60

WORKED SOLUTION & VERIFICATION

The percentage explained is $\frac{51}{51+49} \times 100\% = 51\%$.

11 MCQ

Consider an i.i.d. sample $x_1, x_2, \ldots, x_n$ from a Poisson distribution with probability mass function $P(X = k; \theta) = \frac{\theta^k e^{-\theta}}{k!}$ for $k = 0, 1, 2, \ldots$, where $\theta > 0$. Given a sample of size $n = 5$ with sample sum $\sum_{i=1}^{5} x_i = 15$, find the maximum likelihood estimate (MLE) $\hat{\theta}$ of θ.

A 5

B 15

C 2

D 3 ✓

WORKED SOLUTION & VERIFICATION

The log-likelihood function for n i.i.d. Poisson(\/θ) observations is:

$$\ell(\theta) = \sum_{i=1}^{n} x_i \ln \theta - n\theta - \sum_{i=1}^{n} \ln(x_i!).$$

Setting the derivative with respect to θ equal to zero gives:

$$\frac{d\ell}{d\theta} = \frac{\sum_{i=1}^{n} x_i}{\theta} - n = 0 \implies \hat{\theta} = \frac{1}{n} \sum_{i=1}^{n} x_i = \frac{15}{5} = 3.$$

12 MCQ

Consider a dataset where the number of defective items produced by a machine per hour, Y, follows a Poisson distribution with a rate parameter $\lambda = 3$. The probability mass function (PMF) for a Poisson distribution is given by $P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, 1, 2, \ldots$. What is the value of λ in this scenario?

A $\lambda = 1$

B $\lambda = e^{-3}$

C $\lambda = 3$ ✓

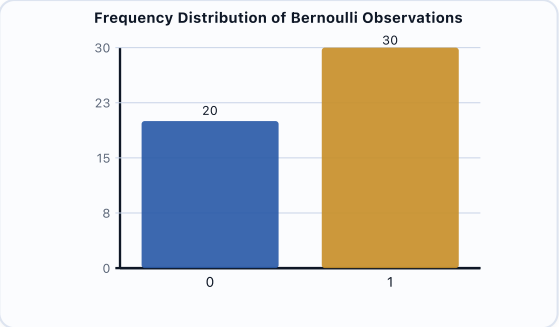
D $\lambda = 0$

WORKED SOLUTION & VERIFICATION

The problem statement explicitly defines that the number of defective items, Y, follows a Poisson distribution with a rate parameter $\lambda = 3$. The rate parameter λ is a direct characteristic of the Poisson distribution that determines its mean and variance. Therefore, the value of λ is directly given as 3.

13 MCQ

A dataset consists of observations $x_1, x_2, \ldots, x_n$ that are assumed to be drawn independently from a Bernoulli distribution with parameter p. The probability mass function is given by $P(X = k; p) = p^k(1-p)^{1-k}$ for $k \in \{0, 1\}$. The following bar chart shows the frequency distribution of these observations, where $n = 50$. Using the method of Maximum Likelihood Estimation (MLE), determine the estimated value of p for this distribution.



A $\frac{3}{5}$ ✓

B $\frac{2}{5}$

C $\frac{1}{2}$

D $\frac{3}{2}$

WORKED SOLUTION & VERIFICATION

The Bernoulli distribution has the probability mass function $P(X = k; p) = p^k(1-p)^{1-k}$. The likelihood function for n independent observations is $L(p; x_1, \ldots, x_n) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i}$. This can be rewritten as $L(p) = p^{\sum x_i}(1-p)^{n-\sum x_i}$.

The log-likelihood function is $\ln L(p) = (\sum x_i) \ln p + (n - \sum x_i) \ln(1-p)$.

To find the maximum likelihood estimate (MLE) of p, we take the derivative of the log-likelihood with respect to p and set it to zero:

$\frac{d}{dp} \ln L(p) = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1-p}$

Setting the derivative to zero:

$\frac{\sum x_i}{p} = \frac{n - \sum x_i}{1-p}$

$(\sum x_i)(1-p) = p(n - \sum x_i)$

$\sum x_i - p\sum x_i = np - p\sum x_i$

$\sum x_i = np$

$\hat{p}_{MLE} = \frac{\sum x_i}{n}$

This means the MLE for p is the sample mean.

From the bar chart, the frequency of observations equal to 0 is 20, and the frequency of observations equal to 1 is 30. The total number of observations is $n = 50$.

The sum of the observations $\sum x_i$ is calculated as:

$\sum x_i = (20 \times 0) + (30 \times 1) = 0 + 30 = 30$

Therefore, the MLE for p is:

$\hat{p}_{MLE} = \frac{30}{50} = \frac{3}{5}$

Common algebraic traps: Students might incorrectly calculate the sample mean, or confuse the counts of 0s and 1s. The option $\frac{2}{5}$ would arise from incorrectly using $\frac{20}{50}$ which represents $1-p$. The option $\frac{1}{2}$ might arise from assuming equal probabilities without considering the data. The option $\frac{3}{2}$ is not a valid probability.

14 MCQ

Consider a sequence of n independent Bernoulli trials, where the probability of success in each trial is p. Let X be the random variable representing the number of successes in these n trials. Thus, $X \sim \text{Binomial}(n, p)$. We are given a dataset of N observed values $\{x_1, x_2, \ldots, x_N\}$ for X, where each x_i is an observed count of successes from n trials. The observed data are: [3, 4, 2, 5, 4, 3, 2, 4, 5, 3]. If we assume that $n = 5$ (the number of trials for each observation is fixed at 5), find the value of p that maximizes the likelihood of observing this dataset. Then, compute the maximum likelihood estimate (MLE) of the probability of success, p, and round your answer to two decimal places.

- **A** 0.70 ✓
- **B** 0.75
- **C** 0.60
- **D** 0.67

WORKED SOLUTION & VERIFICATION

The random variable X follows a Binomial distribution: $X \sim \text{Binomial}(n, p)$. The probability mass function (PMF) of the Binomial distribution is given by $P(X = k|n, p) = \binom{n}{k} p^k (1-p)^{n-k}$, where k is the number of successes.

We are given $n = 5$ and a dataset of $N = 10$ observations: $\{x_1, x_2, \ldots, x_{10}\} = \{3, 4, 2, 5, 4, 3, 2, 4, 5, 3\}$.

The likelihood function for this dataset, given p, is the product of the PMFs for each observation:

$L(p|\{x_i\}) = \prod_{i=1}^{N} P(X = x_i|n, p) = \prod_{i=1}^{10} \binom{5}{x_i} p^{x_i} (1-p)^{5-x_i}$

To find the MLE of p, we typically work with the log-likelihood function, which simplifies the product into a sum:

$\ell(p) = \log L(p) = \sum_{i=1}^{10} \log \left(\binom{5}{x_i} p^{x_i} (1-p)^{5-x_i} \right)$

$\ell(p) = \sum_{i=1}^{10} \left(\log \binom{5}{x_i} + x_i \log p + (5 - x_i) \log(1-p) \right)$

$\ell(p) = \sum_{i=1}^{10} \log \binom{5}{x_i} + \left(\sum_{i=1}^{10} x_i \right) \log p + \left(\sum_{i=1}^{10} (5 - x_i) \right) \log(1-p)$

The term $\sum_{i=1}^{10} \log \binom{5}{x_i}$ is a constant with respect to p and can be ignored for maximization.

Let $S = \sum_{i=1}^{10} x_i$ be the total number of successes in the dataset, and $T = \sum_{i=1}^{10} (5 - x_i) = 10 \times 5 - \sum_{i=1}^{10} x_i = 50 - S$ be the total number of failures.

So, $\ell(p) \approx S \log p + T \log(1-p)$.

To find the MLE, we take the derivative of $\ell(p)$ with respect to p and set it to zero:

$\frac{d\ell}{dp} = \frac{S}{p} - \frac{T}{1-p} = 0$

$\frac{S}{p} = \frac{T}{1-p}$

$S(1-p) = Tp$

$S - Sp = Tp$

$S = Tp + Sp$

$S = p(T + S)$

$p = \frac{S}{T+S}$

We know that $T + S = \sum_{i=1}^{10} (5 - x_i) + \sum_{i=1}^{10} x_i = \sum_{i=1}^{10} (5 - x_i + x_i) = \sum_{i=1}^{10} 5 = 10 \times 5 = 50$.

Now, let's calculate S, the sum of the observed successes:

$S = 3 + 4 + 2 + 5 + 4 + 3 + 2 + 4 + 5 + 3 = 35$.

Therefore, the MLE of p is:

$\hat{p} = \frac{S}{nN} = \frac{35}{5 \times 10} = \frac{35}{50} = \frac{7}{10} = 0.70$.

The maximum likelihood estimate of p is 0.70.

Let's check the second derivative to confirm it's a maximum:

$\frac{d^2\ell}{dp^2} = -\frac{S}{p^2} - \frac{T}{(1-p)^2}$. Since $S > 0$, $T > 0$, $p \in (0, 1)$, the second derivative is negative, confirming a maximum.

Common Misconception:

A common mistake might be to incorrectly calculate the sum of successes or failures, or to use the wrong formula for the MLE of a binomial proportion, such as simply averaging the proportions of successes in each trial (which is not applicable here as n is fixed). For example, if one made an arithmetic error and got $S = 30$, then $\hat{p} = 30/50 = 0.60$. If $S = 37.5$ (an impossible sum, but for illustration of error), $\hat{p} = 37.5/50 = 0.75$. If $S = 30$, $\hat{p} = 30/50 = 0.6$. If $S = 33.5$, $\hat{p} = 33.5/50 = 0.67$.

15 MCQ

A binary classifier $f(x)$ produces the confusion matrix below with $\text{TP} = 80$, $\text{FP} = 10$, $\text{FN} = 42$, and $\text{TN} = 65$. Determine the total number of correct classifications $g = \text{TP} + \text{TN}$.

A 165	B 149
C 140	D **145** ✓

WORKED SOLUTION & VERIFICATION

Summing $\text{TP} + \text{TN} = 80 + 65 = 145$.

16 MCQ

A dataset consists of three independent observations X_1, X_2, X_3 drawn from a Poisson distribution with parameter λ. The observed values are $x_1 = 3, x_2 = 5, x_3 = 4$. We want to find the Maximum Likelihood Estimate (MLE) for λ. The log-likelihood function for a Poisson distribution is given by $\ell(\lambda) = \sum_{i=1}^{n} (x_i \log(\lambda) - \log(x_i!) - \lambda)$.

Part 1: Derive the expression for the derivative of the log-likelihood function with respect to λ.

Part 2: Set the derivative to zero to find the MLE of λ.

A $\hat{\lambda} = \frac{1}{3}$	B $\hat{\lambda} = \frac{3+5+4}{3} = 4$ ✓
C $\hat{\lambda} = \frac{3+5+4}{3 \times e}$	D $\hat{\lambda} = \frac{\sum x_i}{\sum \log(x_i!)}$

WORKED SOLUTION & VERIFICATION

The log-likelihood function for $n = 3$ observations is $\ell(\lambda) = (x_1 \log(\lambda) - \log(x_1!) - \lambda) + (x_2 \log(\lambda) - \log(x_2!) - \lambda) + (x_3 \log(\lambda) - \log(x_3!) - \lambda)$.

Part 1: The derivative of the log-likelihood function with respect to λ is:

$\frac{d\ell}{d\lambda} = \sum_{i=1}^{3} (\frac{x_i}{\lambda} - 1) = (\frac{x_1}{\lambda} - 1) + (\frac{x_2}{\lambda} - 1) + (\frac{x_3}{\lambda} - 1)$.

Part 2: To find the MLE, we set the derivative to zero:

$\frac{x_1}{\lambda} - 1 + \frac{x_2}{\lambda} - 1 + \frac{x_3}{\lambda} - 1 = 0$

$\frac{x_1+x_2+x_3}{\lambda} - 3 = 0$

$\frac{x_1+x_2+x_3}{\lambda} = 3$

$\hat{\lambda} = \frac{x_1+x_2+x_3}{3}$.

Substituting the observed values $x_1 = 3, x_2 = 5, x_3 = 4$:

$\hat{\lambda} = \frac{3+5+4}{3} = \frac{12}{3} = 4$.

Common algebraic traps include incorrectly differentiating the log-likelihood or making errors in solving for λ. For example, forgetting the -1 term from the derivative of $-\lambda$ leads to an incorrect denominator. Choice B is incorrect as it includes e which is not part of the Poisson MLE derivation. Choice C is an incomplete and incorrect expression. Choice D is only correct if the sum of observations was 12, but the formula for $\hat{\lambda}$ is the sample mean. Therefore, the MLE is $\hat{\lambda} = 4$.

17 MCQ

From the given relation $n = 8$ independent observations from a normal distribution with sum $\sum_{i=1}^{8} x_i = 24$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

- A $\hat{\mu}_{\text{MLE}} = 6$
- B $\hat{\mu}_{\text{MLE}} = 3$ ✓
- C $\hat{\mu}_{\text{MLE}} = 24$
- D $\hat{\mu}_{\text{MLE}} = 1.5$

WORKED SOLUTION & VERIFICATION

The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{24}{8} = 3$.

18 MCQ

Provided with $n = 12$ independent observations from a normal distribution with sum $\sum_{i=1}^{12} x_i = 36$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

- A $\hat{\mu}_{\text{MLE}} = 6$
- B $\hat{\mu}_{\text{MLE}} = 36$
- C $\hat{\mu}_{\text{MLE}} = 3$ ✓
- D $\hat{\mu}_{\text{MLE}} = 1.5$

WORKED SOLUTION & VERIFICATION

The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{36}{12} = 3$.

19 MCQ

Based on the condition $n = 16$ independent observations from a normal distribution with sum $\sum_{i=1}^{16} x_i = 48$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

- A $\hat{\mu}_{\text{MLE}} = 3$ ✓
- B $\hat{\mu}_{\text{MLE}} = 6$
- C $\hat{\mu}_{\text{MLE}} = 48$
- D $\hat{\mu}_{\text{MLE}} = 1.5$

WORKED SOLUTION & VERIFICATION

The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{48}{16} = 3$.

20 MCQ

Starting from $n = 20$ independent observations from a normal distribution with sum $\sum_{i=1}^{20} x_i = 60$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

- A $\hat{\mu}_{\text{MLE}} = 6$
- B $\hat{\mu}_{\text{MLE}} = 60$
- C $\hat{\mu}_{\text{MLE}} = 3$ ✓
- D $\hat{\mu}_{\text{MLE}} = 1.5$

WORKED SOLUTION & VERIFICATION

The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{60}{20} = 3$.

21 MCQ

Knowing that $n = 24$ independent observations from a normal distribution with sum $\sum_{i=1}^{24} x_i = 72$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

- **A** $\hat{\mu}_{\text{MLE}} = 3$ ✓
- **B** $\hat{\mu}_{\text{MLE}} = 6$
- **C** $\hat{\mu}_{\text{MLE}} = 72$
- **D** $\hat{\mu}_{\text{MLE}} = 1.5$

WORKED SOLUTION & VERIFICATION

The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{72}{24} = 3$.

22 MCQ

Considering training a language model token predictor, a binary token occurs $k = 6$ times across $n = 20$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

- **A** $\hat{p}_{\text{MLE}} = 0.450$
- **B** $\hat{p}_{\text{MLE}} = 0.300$ ✓
- **C** $\hat{p}_{\text{MLE}} = 0.150$
- **D** $\hat{p}_{\text{MLE}} = 0.700$

WORKED SOLUTION & VERIFICATION

For a Bernoulli likelihood $L(p) = p^k(1-p)^{n-k}$, the log-likelihood is $\ln L = k\ln p + (n-k)\ln(1-p)$. Setting derivative to zero: $\frac{k}{p} - \frac{n-k}{1-p} = 0 \implies \hat{p} = \frac{k}{n} = \frac{6}{20} = 0.300$.

23 MCQ

Examining training a language model token predictor, a binary token occurs $k = 9$ times across $n = 30$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

- **A** $\hat{p}_{\text{MLE}} = 0.450$
- **B** $\hat{p}_{\text{MLE}} = 0.300$ ✓
- **C** $\hat{p}_{\text{MLE}} = 0.150$
- **D** $\hat{p}_{\text{MLE}} = 0.700$

WORKED SOLUTION & VERIFICATION

For a Bernoulli likelihood $L(p) = p^k(1-p)^{n-k}$, the log-likelihood is $\ln L = k\ln p + (n-k)\ln(1-p)$. Setting derivative to zero: $\frac{k}{p} - \frac{n-k}{1-p} = 0 \implies \hat{p} = \frac{k}{n} = \frac{9}{30} = 0.300$.

24 MCQ

For the configuration in training a language model token predictor, a binary token occurs $k = 12$ times across $n = 40$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

- **A** $\hat{p}_{\text{MLE}} = 0.450$
- **B** $\hat{p}_{\text{MLE}} = 0.150$
- **C** $\hat{p}_{\text{MLE}} = 0.300$ ✓
- **D** $\hat{p}_{\text{MLE}} = 0.700$

WORKED SOLUTION & VERIFICATION

For a Bernoulli likelihood $L(p) = p^k(1-p)^{n-k}$, the log-likelihood is $\ln L = k\ln p + (n-k)\ln(1-p)$. Setting derivative to zero: $\frac{k}{p} - \frac{n-k}{1-p} = 0 \implies \hat{p} = \frac{k}{n} = \frac{12}{40} = 0.300$.

25 MCQ

Analyzing training a language model token predictor, a binary token occurs $k = 15$ times across $n = 50$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

- **A** $\hat{p}_{\text{MLE}} = 0.300$ ✓
- **B** $\hat{p}_{\text{MLE}} = 0.450$
- **C** $\hat{p}_{\text{MLE}} = 0.150$
- **D** $\hat{p}_{\text{MLE}} = 0.700$

WORKED SOLUTION & VERIFICATION

For a Bernoulli likelihood $L(p) = p^k(1-p)^{n-k}$, the log-likelihood is $\ln L = k \ln p + (n-k)\ln(1-p)$. Setting derivative to zero: $\frac{k}{p} - \frac{n-k}{1-p} = 0 \implies \hat{p} = \frac{k}{n} = \frac{15}{50} = 0.300$.